Student's Name:.....

Signature.....

Random No.						Personal No.		

BIOLOGY
PAPER 1
(Theory)
2 Hours 30 minutes

SELF-STUDY: BIOCLASS UG

END OF CYCLE EXAMINATIONS TERM III 2025

BIOLOGY

Paper 1

(Theory)

2 HOURS 30 Minutes

INSTRUCTTIONS TO CANDIDATES:

This paper consists of **two** *sections:* **A** *and* **B.** *It has* **FIVE** *examination items.*

Section **A** *has* **Two Compulsory** *items.*

Section **B** *has* **two** *ITEMS: Answer* **one ITEM**.

Answers to section **A must** *be written in the spaces provided while answers to* **Section B** *must be written in the answer booklet(s) provided.*

Answer **THREE** *items in all.*

Any additional item(s) answered will **not** *be scored*

ITEM 1

In a Southeast Asian coastal village, illegal cyanide fishing is practiced fishermen capture live fish for aquariums. This has led to a significant die-off of non-target coral reef species. A marine biologist, Dr. Lim, collects tissue samples from affected damselfish. Cyanide is a potent inhibitor of cytochrome c oxidase.

Table: Metabolic Analysis of Damselfish Gill Tissue

Sample	Cytochrome c Oxidase Activity (% of normal)	Gill Cell Membrane Integrity (% of cells damaged)	Gill ATP Concentration (mM)	Oxygen Uptake Rate (mg O_2/g/h)
Unaffected Fish	100	5	4.8	9.5
Cyanide-Exposed Fish	3	60	0.8	1.2

Task:

(a) Explain, in simple steps, how cyanide inhibition of cytochrome c oxidase leads to the changes shown in Table 1.

(b) Using the information provided, propose strategies to solve the challenges.

ITEM 2

In a school greenhouse, maize (C_4 plant) developed yellow patches and stunted growth under prolonged artificial light. Temperature fluctuated widely and Carbon dioxide concentration was low. Microscopy revealed partial loss of Kranz anatomy and reduced PEP carboxylase activity.

Condition	Light (hrs/day)	Temperature Range (°C)	CO_2 (ppm)	PEP Carboxylase (µmol CO_2/mg/hr)	Rubisco (µmol CO_2/mg/hr)	Starch (mg/g)	Protein (% dry mass)	Lipid (% dry mass)	Chlorophyll (mg/g)	Guard Cell Aperture (µm)
Optimal	12	25–28	400	7.8	5.4	25	14	8	2.8	4.2
Observed	20	38/15	220	2.3	1.9	10	6	3	1.2	1.2

Task

(a) (i) Analyze how light duration, temperature fluctuation, and CO_2 concentration collectively affect C_4 enzyme activities and chloroplast structure.

(ii) Explain how changes in these variables influence macromolecule synthesis (starch, proteins, lipids) and stomatal regulation.

(b) Propose integrated strategies to optimize each factor and restore healthy photosynthesis and growth.

SECTION B

Attempt one item

ITEM 3

In a busy Kampala hospital, patients in the COVID-19 recovery ward were monitored for oxygen levels and immune response. Three patients (Rita, Sam, and Fiona) were studied: Fiona developed slight facial swelling after the second vaccine dose but recovered quickly. Sam developed mild dehydration symptoms and elevated heart rate.

Parameter	Rita	Sam	Fiona
Vaccination status	Fully vaccinated	Not vaccinated	Fully vaccinated
Blood oxygen (%)	96	87	90
Antibody titre (AU/mL)	1000	40	620
White blood cell count ($\times 10^9$/L)	7.5	10.0	5.5
Temperature (°C)	36.9	38.4	37.2
ADH level (pg/mL)	7	15	6

Reference: *"Immune Response and Homeostatic Balance in COVID-19 Recovery" Nature Immunology (2022).*

Task:
(a) Explain how immunity and homeostasis account for the differences in recovery and physiological data among the three patients.
(b) Describe how antibodies and ADH helped Rita and Fiona maintain stability compared to Sam.
(c) Propose safe management strategies to enhance recovery in similar patients.

ITEM 4
At Kisubi High School, the Biology teacher organized a health awareness week. Students learned about the importance of vaccination, good nutrition, personal hygiene, and physical fitness. During one session, a demonstration was done to compare immune responses between vaccinated and unvaccinated students after exposure to a harmless simulated pathogen. After the activity, teachers encouraged students to maintain healthy habits such as eating well, keeping dormitories clean, getting enough rest, and taking part in morning exercises to boost their general body immunity.

Observation	Vaccinated Students	Unvaccinated Students
Body temperature (°C)	37.0	38.2
White blood cell count (relative)	High	Moderate
Presence of antibodies	Present	Absent
Reaction time to infection (days)	1	5

Reference: *"Principles of Immunity and Vaccination, Secondary Biology Digest (2024)"*
Task
(a) Explain the difference between innate and adaptive immunity shown in this demonstration.
(b) Describe how vaccination prepares the body to respond quickly during infection.
(c) Suggest school-level health measures that support immunity among students.

TAP THE LINK BELOW FOR THE SOLUTIONS TO THIS PAPER
https://www.youtube.com/watch?v=hAGrgppGh0o

END